Q **August 4, 2026** Elton Liu

[ERR] Plasma Cannon Launcher **[13 pts]**

Kai'Sa, the Daughter of the Void, fires plasma bolts from a device that can be modelled as an *elastic rotating launcher*. While charging a shot, Kai'Sa spins up a lightweight rigid launch arm of length L (the arm's own mass is negligible); a plasma bolt of mass m is fixed to the arm's tip.

During the charging phase the arm rotates counter-clockwise about a fixed pivot, starting from rest at the horizontal. A variable external torque turns the arm against a torsional "Hextech" elastic structure, which produces a restoring torque

$$\tau = -k\theta,$$

where θ is the arm's angle from the horizontal. When the arm reaches a maximum angle θ_0, the bolt is released and flies off along the tangent, undergoing projectile motion under gravity g. Air resistance, friction, the arm's mass, and all energy losses are neglected.

Given quantities: m, L, k, θ_0, g.

Assumptions

1. The arm is rigid and massless; the bolt is a point particle.
2. The torsional spring obeys the linear law $\tau = -k\theta$.
3. During charging, gravity's torque on the bolt is neglected (it is reintroduced only in Q5).

Questions.

Q1. Work in the charging phase. Find the total work done by the external torque in charging the arm from $\theta = 0$ to $\theta = \theta_0$ against the torsional spring.

Q2. Rotational energy and angular speed. At the instant of release, find

i. the system's angular speed ω;
ii. the bolt's instantaneous speed v;
iii. the bolt's instantaneous centripetal acceleration a_c.

Q3. Angular momentum: conserved or not? A student claims: "from the charging rotation through the instant of release, the system's angular momentum about the pivot is conserved." Judge whether this is right or wrong, and give a rigorous physical argument using torque and the angular-momentum theorem.

Q4. Projectile motion and mechanical energy. The launch point is at height H, and the launch angle is θ_0 (the velocity points along the tangent, at angle θ_0 above the horizontal). The bolt flies off and lands on the ground at height 0.

i. Using conservation of mechanical energy, write an expression for the landing speed v_f (no need to work out the kinematics).
ii. If θ_0 is very small (small-angle approximation), explain briefly how the horizontal range changes as θ_0 increases, and give the physical reason.

Q5. Model extension: critical equilibrium. Now include the torque that gravity exerts on the bolt. Find the equilibrium angle position(s) θ at which the bolt can remain in static equilibrium (net torque zero).